

EXHIBIT 38

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U.S. Patent No. 10,062,748 Infringement Chart

Title: Segmented guard-ring and chip edge seals

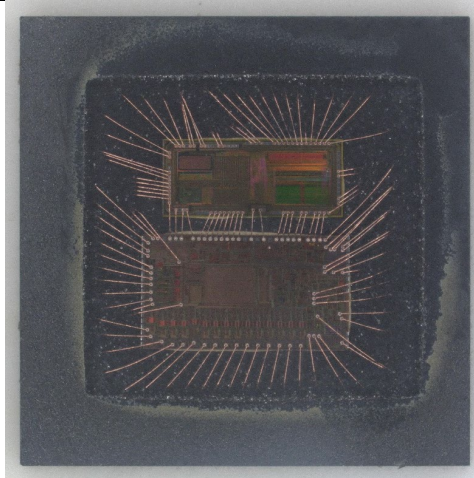
Inventors: Anthony K. Stamper, Vincent J. McGahay, Zhong-Xiang He

Abstract: The present disclosure relates to semiconductor structures and, more particularly, to segmented guard-ring and chip edge seals and methods of manufacture. The structure includes: a guard ring structure formed in a low-k dielectric material; and an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material.

Claim	Tower '748 Accused Products
[1pre] A structure comprising:	The '748 Accused Tower Products include products manufactured using Tower's BCD processes (including its TS18SOI process).¹ For example, Tower manufactures the PA22BZ die in the Qorvo PAC22140 using a 180nm BCD process referred to by Tower as TS18SOI. To the extent the preamble is limiting, the Tower '983 Accused Products comprise a structure, as explained further in the following limitations.  On information and belief, the PA22BZ (bottom of image below) has substantially similar structure, function, operation, and implementation with respect to the claims of the '748 patent as any other Tower '748 Accused Products.

¹ GlobalFoundries reserves the right to identify different or additional Tower '748 Accused Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

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For example, a cross-sectional image of the PA22BZ die (as illustrated by the red line in the image below) reveals a structure that is representative of the infringing functionality in the '748 Tower Accused Products.

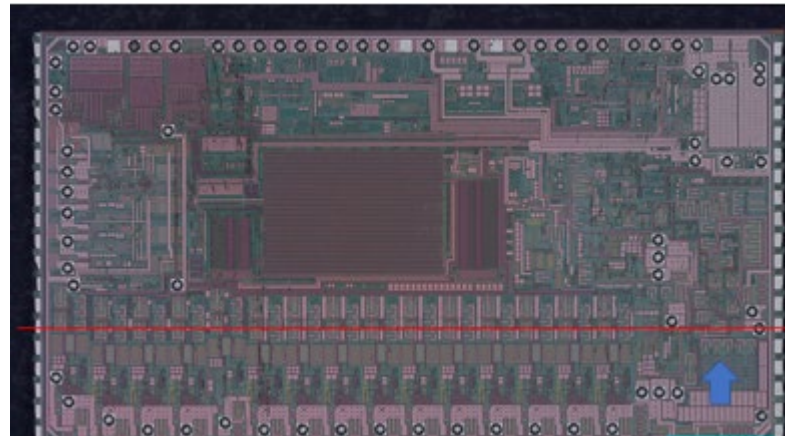
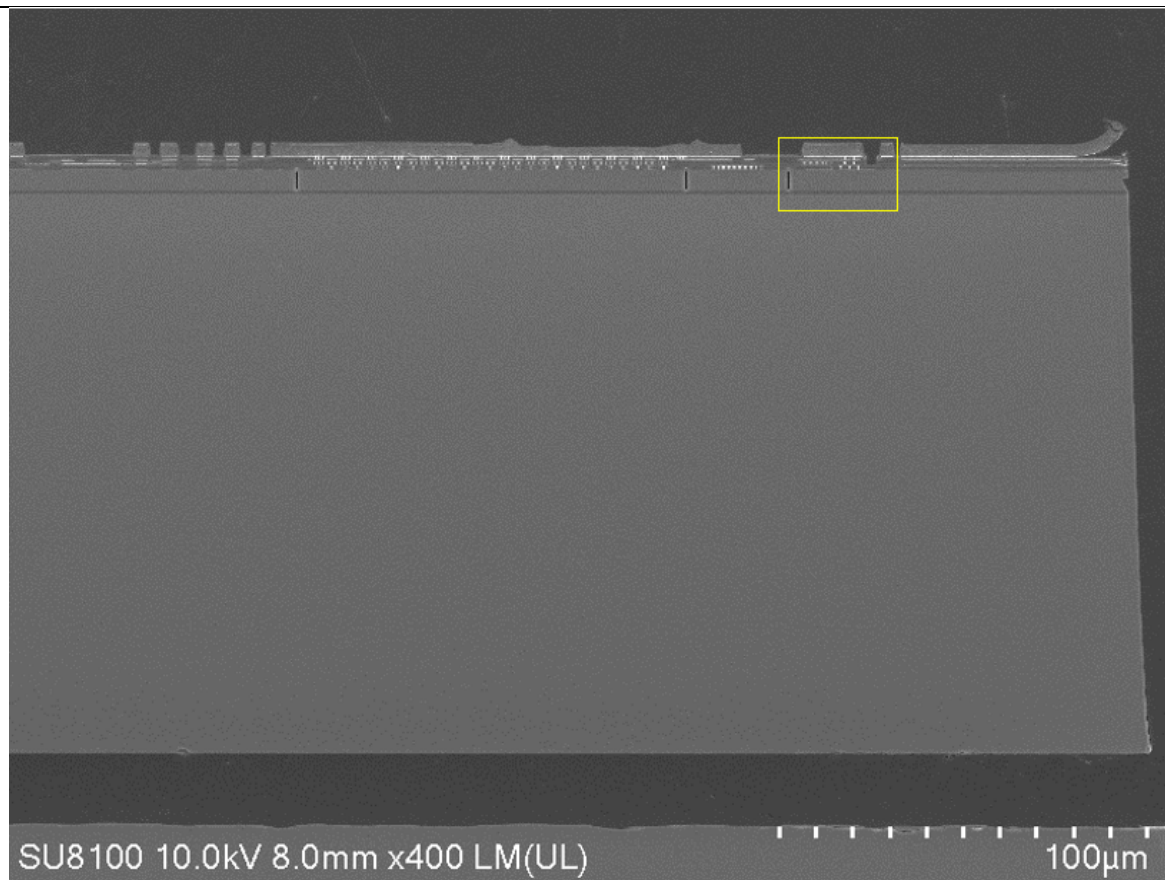


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For example, the Tower '748 Accused Products include a structure (contained in the yellow box in the image above).

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[1a] a guard ring structure formed in a low-k dielectric material; and

The Tower '748 Accused Products include a guard ring structure that is formed in a material that has a low dielectric constant.

For example, a cross-sectional image of the PA22BZ die reveals a guard ring structure formed in a low-k dielectric material that, on information and belief, is representative of the guard rings used in the Tower '748 Accused Products.

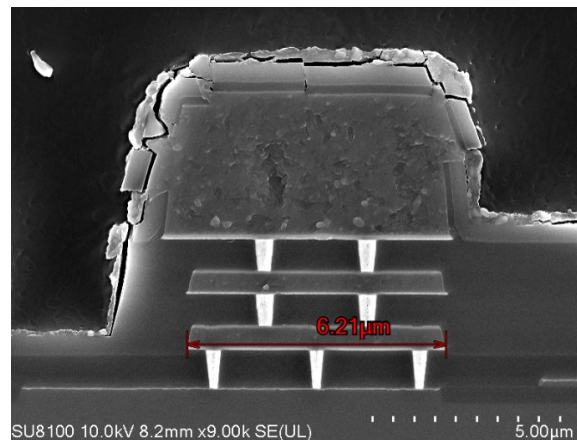
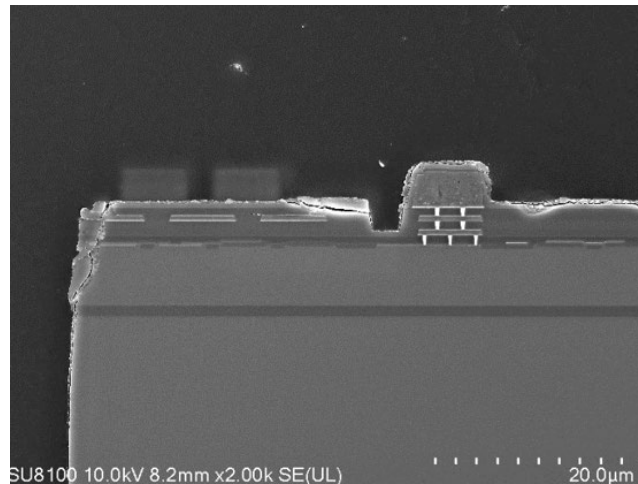


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As shown in the images below, the guard ring structure is formed in a low-k dielectric material.

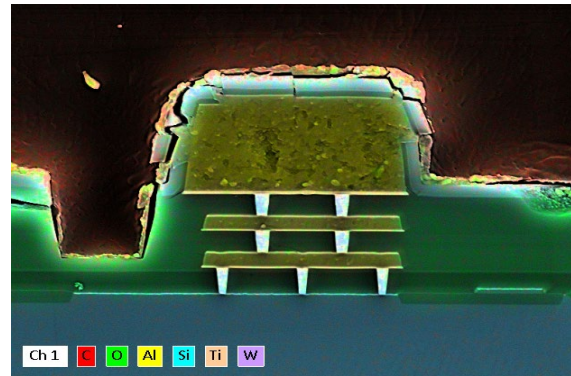


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[1b] an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material,

The Tower '748 Accused Products comprise an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material.

For example, in the yellow box in the image below, the edge seal structure (left side of yellow box) is formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material.

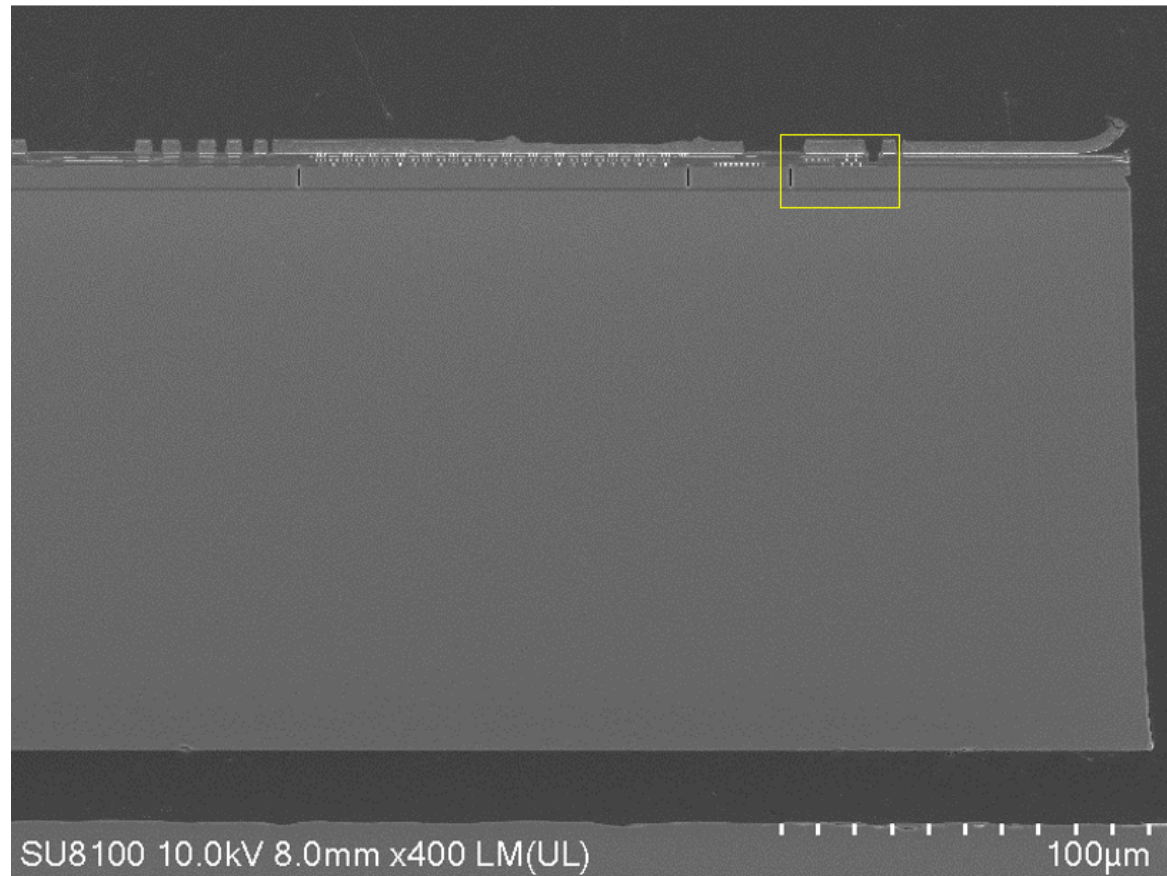


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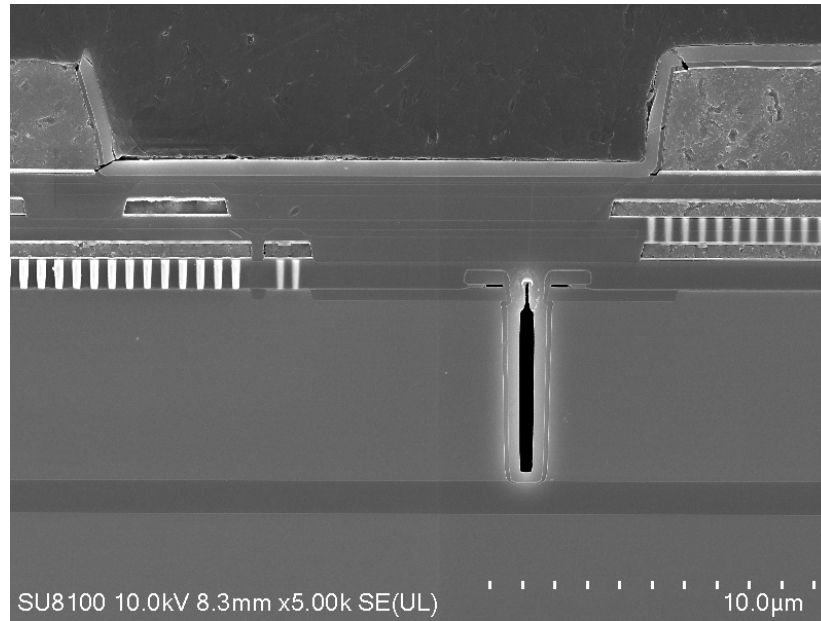


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[1c] wherein the substrate is a silicon on insulator substrate and the edge seal structure terminates below the low-k dielectric material.

The Tower '748 Accused Products comprise a silicon on insulator substrate and the edge seal structure terminates below the low-k dielectric material.

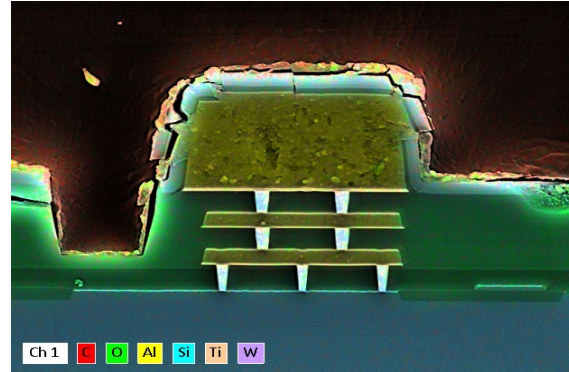
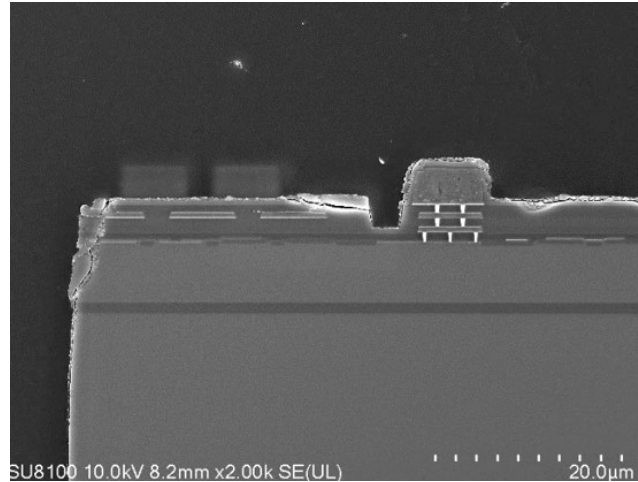


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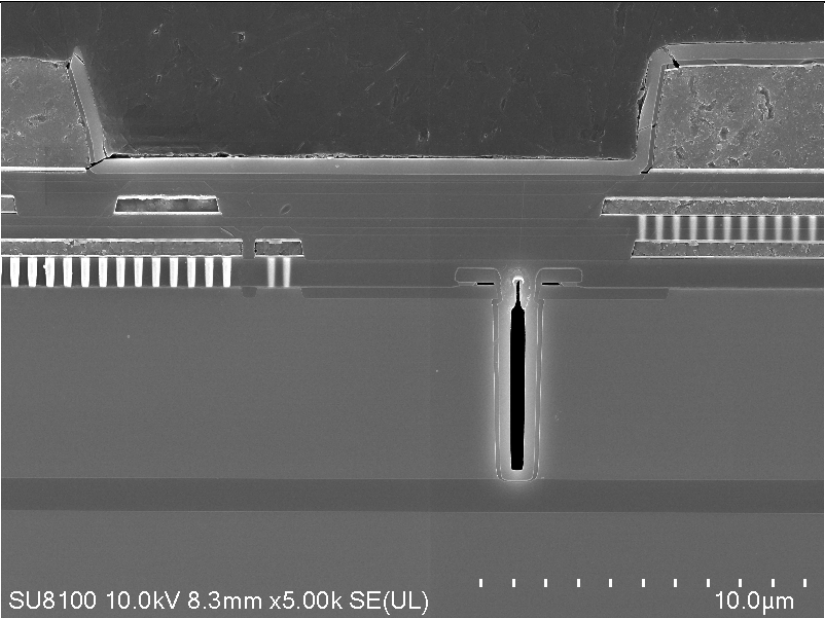
	
[9pre] A structure comprising:	To the extent the preamble is limiting, the Tower '748 Accused Products comprise a structure. <i>See</i> [1pre].
[9a] a guard ring structure formed in a low-k dielectric material; and	The Tower '748 Accused Products comprise a guard ring structure formed in a low-k dielectric material. <i>See</i> [1a].
[9b] an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material,	The Tower '748 Accused Products comprise an edge seal structure formed through the low-k dielectric material to at least a substrate underneath the low-k dielectric material. <i>See</i> [1b].

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[9c] wherein the edge seal structure is one or more airgaps formed in the low-k dielectric material that is pinched off with oxide.

In the Tower '748 Accused Products, the edge seal structure is one or more airgaps formed in the low-k dielectric material that is pinched off with oxide, as shown below.

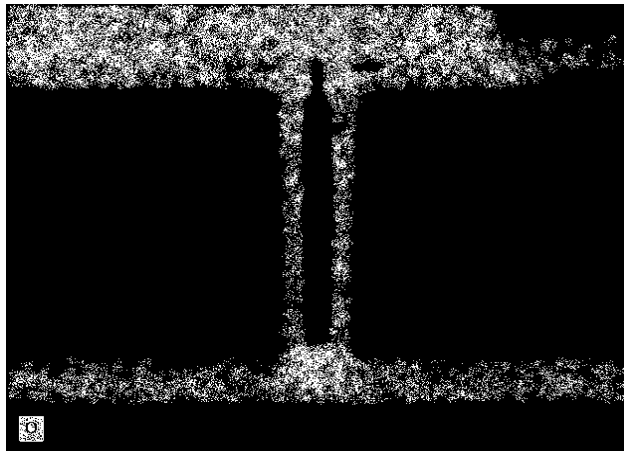
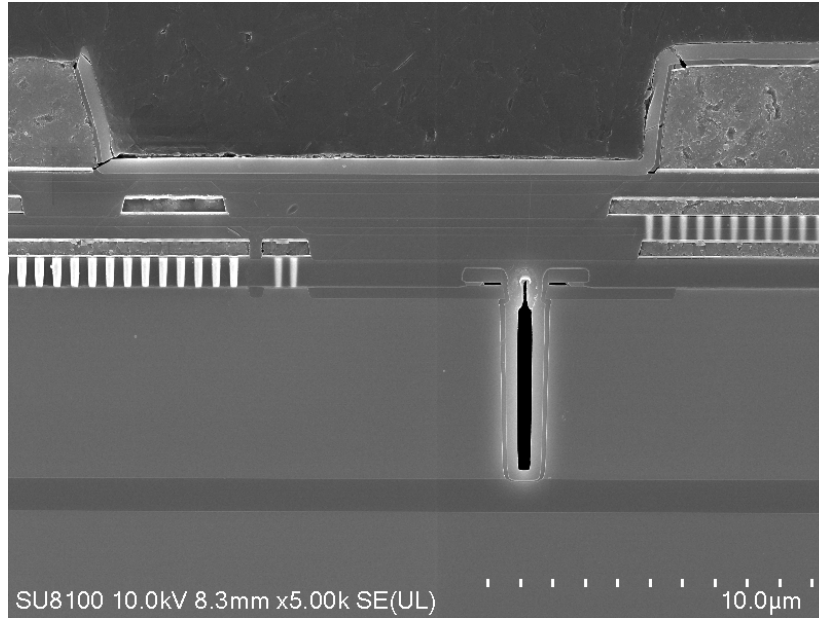


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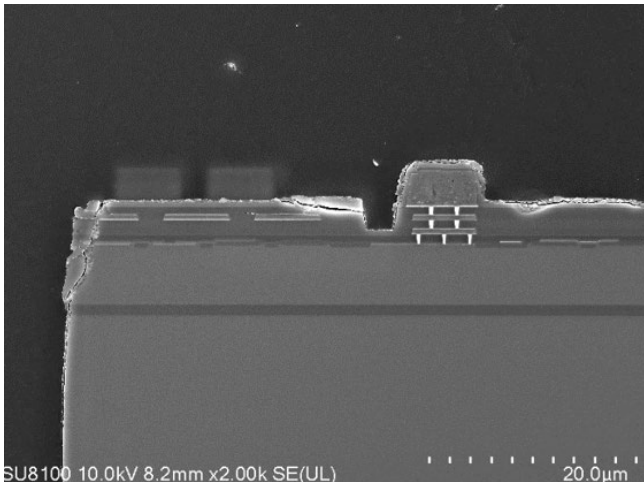
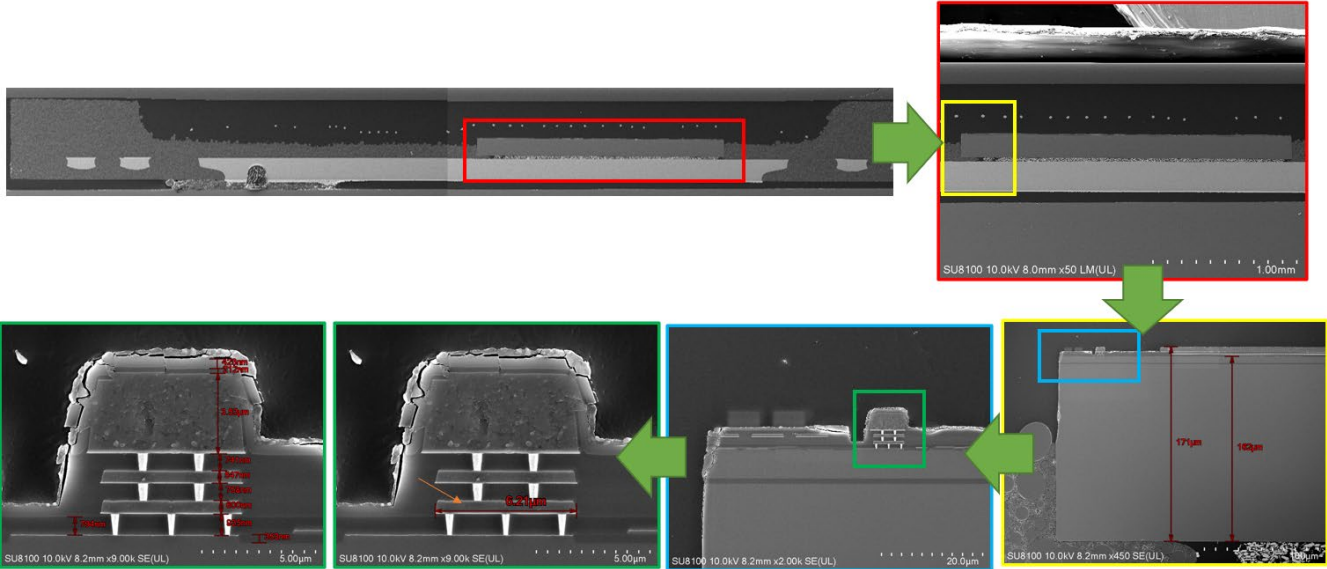
[11pre] A structure comprising:	To the extent the preamble is limiting, the Tower '748 Accused Products comprise a structure. <i>See</i> [1pre].
[11a] a silicon on insulator substrate;	The Tower '748 Accused Products comprise a silicon on insulator substrate. <i>See</i> [1c].
[11b] a middle of the line and back end of the line dielectric low-k dielectric material on the silicon on insulator substrate;	The Tower '748 Accused Products comprise a middle of the line and back end of the line dielectric low-k dielectric material on the silicon on insulator substrate. <i>See</i> [1a].
[11c] a wiring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material to the silicon on insulator substrate;	The Tower '748 Accused Products comprise a wiring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material to the silicon on insulator substrate. 

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[11d] a segmented guard ring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material; and	The Tower '748 Accused Products comprise a segmented guard ring structure formed in the middle of the line and back end of the line dielectric low-k dielectric material. 
[11e] an edge seal structure formed extending through the middle of the line and back end of the line dielectric low-k dielectric material.	The Tower '748 Accused Products comprise an edge seal structure formed extending through the middle of the line and back end of the line dielectric low-k dielectric material. <i>See [1b].</i>